

77. THE % OF VO₂MAX AT LACTATE THRESHOLD DOESN'T DISCRIMINATE THE PERFORMANCE LEVEL

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A widespread belief is that higher level athletes have a second threshold/FTP at a higher percentage of the maximum oxygen consumption (VO₂max).

Is it really true?

Having a second threshold/FTP at a higher percentage of VO₂max is one of the factors that distinguish elite athletes from non-elite ones?

A hot off the press study published by Johansen and Colleagues (University of South-Eastern Norway) on *International Journal of Sport Physiology and Performance* in 2025 tried to answer this question.

WHAT DID THEY DO?

The lab metabolic incremental tests of 292 aerobic endurance athletes (212 Males and 80 females; 75 long-distance runners, 23 road cyclists, and 94 cross country skiers) were analysed. Athletes were divided into 3 different performance level groups:

1. **Elite Athletes** (N=71): who were competing at international level.
2. **National Athletes**: (N=158) who were competing at national level.
3. **Regional Athletes**: (N=63) who were competing in regional recreational competitions.

The VO₂max and the % of VO₂max at which second lactate threshold occurred were compared across the 3 different performance level groups.

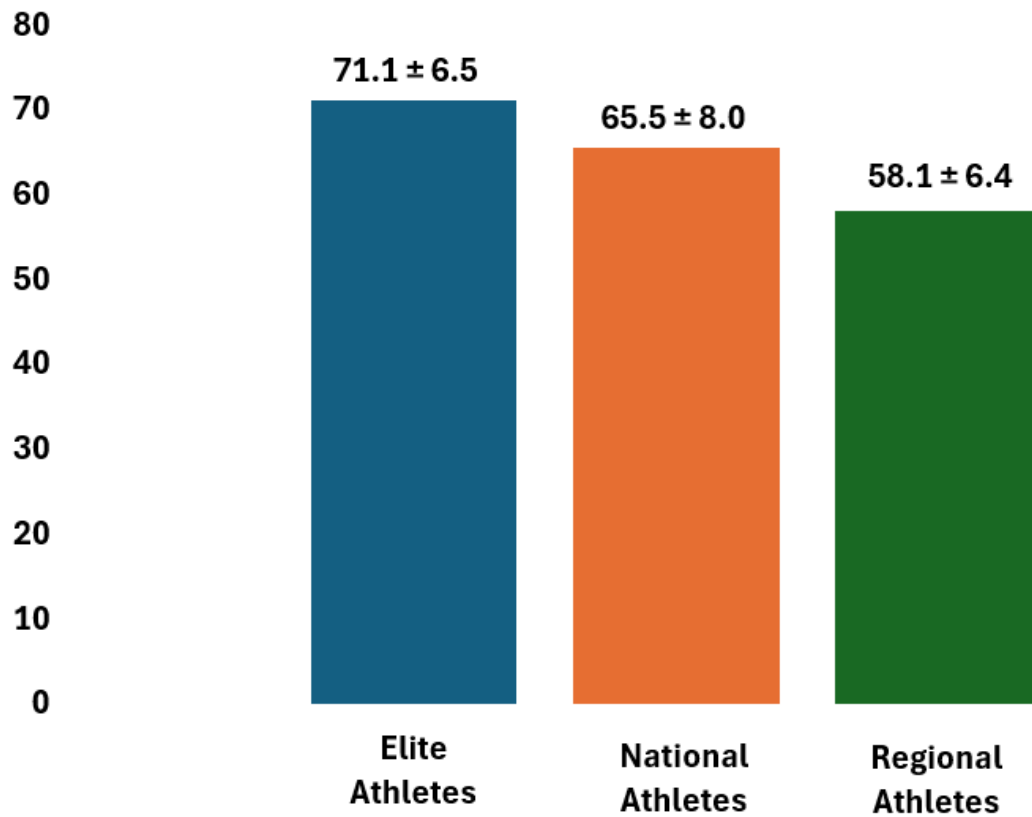
WHAT DID THEY FIND?

As was easily guessed, VO₂max differed significantly between performance levels:

- Elite Athletes: 71.1 ± 6.5 ml/kg/min
- National Athletes: 65.5 ± 8.0 ml/kg/min
- Regional Athletes: 58.1 ± 6.4 ml/kg/min

This was also observed when males and females were analysed separately.

VO₂max (ml/min/kg)



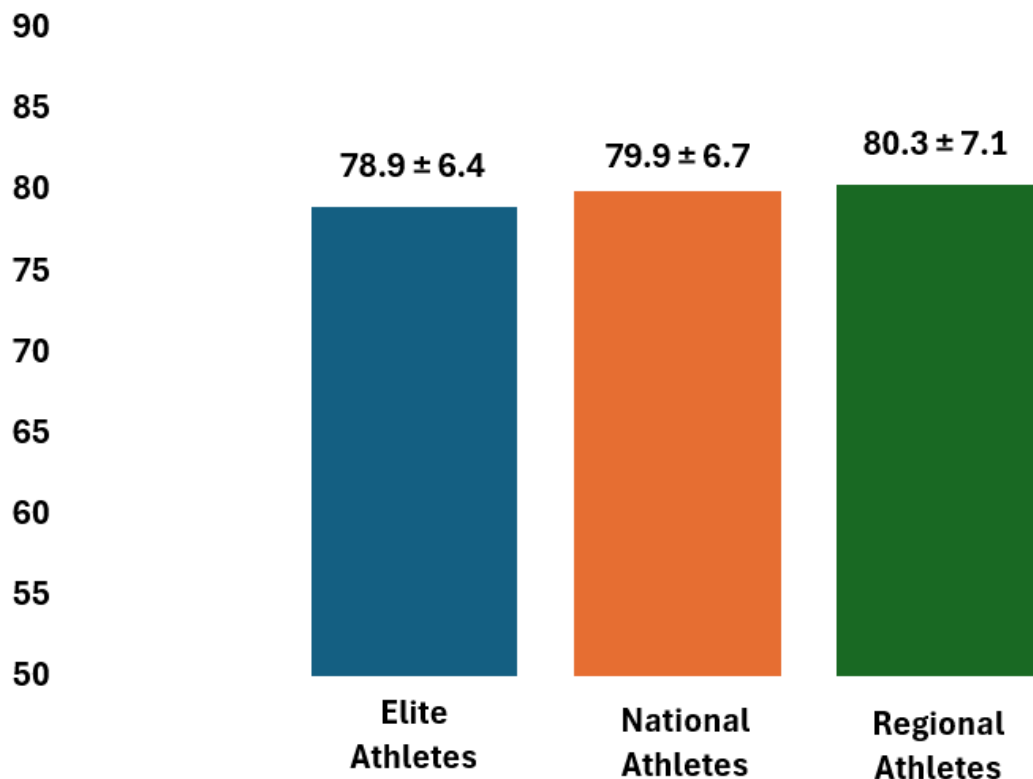
Data from Johansen et al. 2025 (1)

However, interestingly, the % of VO₂max at which second lactate threshold occurred was similar between the three groups:

- Elite Athletes: 78.9 ± 6.4%,
- National Athletes: 79.9 ± 6.7%
- Regional Athletes: 80.3 ± 7.1%

This was also observed when males and females were analysed separately.

Second Lactate Threshold (% of VO₂max)



Data from Johansen et al. 2025 (1).

PRACTICAL TAKEAWAYS:

While having a higher VO₂max is a factor that discriminates the competitive level of endurance athletes, the % of VO₂max at which you have the second lactate threshold is not.

GABRIELE'S COMMENT

I found these data very interesting. They seem to suggest us that **VO₂max discriminates endurance performance level, while the % of VO₂max at lactate threshold doesn't.**

Of course, **this should not be read as 'the % of VO₂max at which you have lactate threshold doesn't impact endurance performance'**. In fact when having similar VO₂max, having FTP at 90% instead of 80% of VO₂max, clearly benefits performance. These data suggest just that (when considered alone) this doesn't discriminate being or not an elite athlete.

Overall, this study can also question that, in already trained athletes, the % of VO₂max at lactate threshold can be raised through training: in fact, despite their higher training volume elite athletes didn't record a second lactate threshold at a higher % of FTP compared to lower performance level athletes.. As the authors commented, this is in line with previously published data which showed that, after a training program, in already trained endurance athletes, similar improvements in VO₂max and power at second lactate threshold were observed, which in turn

resulted in a unchanged % of vo2max at second threshold . From a physiological perspective, this makes sense: endurance training target mainly adaptations of the systems which deliver oxygen to the muscles (stroke volume, haemoglobin level, capillarization) and use oxygen in the muscle (mitochondria density). When these systems are enhanced by training, their improvements are likely to benefit both VO2max and the power/velocity at second threshold. So, it is not probably easy to isolate an improvement in threshold but not in VO2max (it should be an adaptation more marked within the muscles and less at cardiovascular level?). However, this is just a speculation and so it will be interesting to see future studies investigating whether % of VO2max at which FTP occur is a trainable (which strategies?) or a genetic characteristic. This can be very relevant to plan training as this physiological parameter could determine what is the best high intensity interval training session to stimulate aerobic adaptations (learn more [here](#))

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Science

REFERENCES:

1. Johansen JM, Støa EM, Sunde A, Rønnestad BR, Helgerud J, Støren Ø. Elite Aerobic Endurance Performance: Is It Really Related to Lactate Threshold Expressed Relative to Peak Oxygen Uptake? Int J Sports Physiol Perform. 2025 Apr 16:1-5.



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